

Report ID: REP-20260829180236-0211e9  
Target File: Live\_Capture\_Wi-Fi.pcap

Compliance: COMPLIANT (Strong IPsec Security Posture)  
Timestamp: Aug 29, 2026, 18:02 UTC

1. EXECUTIVE SECURITY SUMMARY & POSTURE GRADE

**GRADE A+**      Normalized Risk Index: 10 / 100 (LOW RISK)      Status: Verified by TITAN Engine

2. CRYPTOGRAPHIC INTEGRITY & POST-QUANTUM (PQC) READINESS

- Payload Encryption: Enforced (ESP Protocol 50)

- Authentication Header: None (Pure ESP Tunnel)

- Active SPIs: 0xc743e05b, 0xcb68f974
- Post-Quantum Readiness: 85% (QUANTUM-RESISTANT (CNSA 2.0 Symmetric))

- IKE PSK Exposure Risk: LOW (No IKEv1 Aggressive Mode Hash Exposure)

- CNSA 2.0 Compliance: COMPLIANT

3. TRAFFIC TELEMETRY & PROTOCOL DISTRIBUTION MATRIX

| Protocol Layer / Name                  | Packet Count | Traffic Share (%) | Security Classification |
|----------------------------------------|--------------|-------------------|-------------------------|
| ESP Payload Encryption (Protocol 50)   | 38           | 100.0%            | Encrypted (Secure)      |
| Authentication Header AH (Protocol 51) | 0            | 0.0%              | Cleartext / Control     |
| IKE / NAT-T (UDP Port 500 / 4500)      | 0            | 0.0%              | Cleartext / Control     |
| TCP Transport Streams                  | 0            | 0.0%              | Cleartext / Control     |
| UDP Datagrams (Non-VPN)                | 0            | 0.0%              | Cleartext / Control     |
| DNS Resolution Queries                 | 0            | 0.0%              | Cleartext / Control     |
| ICMP Diagnostic Messages               | 0            | 0.0%              | Cleartext / Control     |
| TOTAL CAPTURED TRAFFIC                 | 38           | 100.0%            | Full Network Stream     |

4. ENCRYPTED TRAFFIC ANALYSIS (ETA) & APPLICATION FINGERPRINTING

- Inferred Application: Interactive Shell / VoIP / Command Stream

- Traffic Pattern: Low Latency / Fixed Size Frames

Behavioral Heuristic: Continuous stream of small, fixed-size encrypted frames indicative of interactive management sessions or voice-over-IP.
- ETA Confidence Score: 94.5%

- Burstiness Index: 0.0

5. MITRE ATT&CK ENTERPRISE FRAMEWORK MAPPING

| Technique ID | Technique Name         | Tactic / Domain      | Severity | Audit Finding                   |
|--------------|------------------------|----------------------|----------|---------------------------------|
| M1037        | Filter Network Traffic | Defensive Mitigation | INFO     | 100% of traffic is encapsulated |

6. DEEP SECURITY AUDIT FINDINGS

- [1] ESP Payload Encryption Verified: 38 encrypted frames (100.0% of capture).
- [2] Full Encapsulation: 100% of captured traffic is encapsulated within secure IPsec tunnel.
- [3] Established Tunnel Trace: IKE initial negotiation phase was completed prior to packet capture.
- [4] Zero Cleartext Leakage: No plaintext transport protocols were observed bypassing the VPN tunnel.
- [5] MTU Headroom Optimal: Average packet length is 170 bytes (Max: 170 bytes), avoiding fragmentation.
- [6] Anti-Replay Protection: Sequence tracking verified across 2 active Security Associations (SAs).

7. ACTIONABLE HARDENING ROADMAP & SIEM CONFIGURATION

Step 1: Maintain current cryptographic parameters and schedule regular re-keying intervals (e.g. every 8-24 hours).